ImageNet Performance Change After Learning the New Task

Nir Moyal
Dan Feldman

1 ImageNet Performance Change

The following table presents the change in ImageNet performance after learning the new task. The difference is calculated relative to the pretrained ImageNet baseline:

$$\Delta\text{ImageNet} = \text{ImageNet Accuracy After Learning the New Task} - \text{Baseline ImageNet Accuracy}$$

Table 1: ImageNet Performance Change After Learning the New Task

Experiment	Method	Baseline ImageNet	ImageNet After New Task	Δ ImageNet
ImageNet → CUB	LwF	62.89	59.49	−3.40
ImageNet → CUB	Fine-Tuning	62.89	53.42	−9.47
ImageNet → CUB	Feature Extraction	62.89	62.89	0.00
ImageNet → Scenes	LwF	62.89	59.82	−3.07
ImageNet → Scenes	Fine-Tuning	62.89	48.69	−14.20
ImageNet → Scenes	Feature Extraction	62.89	62.89	0.00

Negative values indicate a reduction in old-task performance caused by catastrophic forgetting. Feature Extraction preserves the original ImageNet accuracy because the pretrained representation remains frozen. Fine-Tuning causes the strongest degradation, while LwF reduces the loss of old-task performance.